Rijeka Methodology

Curve construction, pricing, market and counterparty risk, XVA, FX and digital-asset derivatives, on-chain confirmation

Version 1.2 · November 2026 · github.com/MikoDevedzic/rijeka · Apache-2.0

How to read this document. Every formula below is the one implemented in the open-source code, with the module named in the margin of each section. Where the code departs from textbook practice, or where a simplification is in force, it is stated as a *known limitation* rather than hidden. The Methodology tab of the application exports, for each product, an Excel workbook that rebuilds the default trade with live formulas from the user's own market data next to the engine's value; those workbooks are the executable companion to this text.

Changes in 1.2 (2 November 2026). FX options in every style and the digital-asset options on the same engine (§13); volatility surfaces as risk factors in VaR and stress, and the market-risk and counterparty-credit chapters (§14, §15), which the previous version left to the repository's notes; the open-box replications and the XVA workbook that carries the engine's own Monte Carlo draws (§9). Version 1.1 (1 November 2026) added the QuantLib reference suite (§18).

Contents

1 Scope and principles **2**
1.1 Notation and conventions . . . 3

2 Discount curve construction **3**
2.1 Instruments . . . 3
2.2 Algorithm . . . 3
2.3 Interpolation and extrapolation . . . 3
2.4 Risk bumps . . . 4

3 Schedules and cashflows **4**

4 Linear interest-rate products **5**

5 European swaptions and the volatility surface **5**
5.1 Annuity, forward and the Bachelier price . . . 5
5.2 Volatility tiers . . . 5
5.3 $\beta = 0$ normal SABR . . . 5

6 Caps, floors and collars **6**

7 Bermudan swaptions: Hull–White trinomial tree **6**

8 Hull–White calibration for exposure simulation **7**

9 Exposure, collateral and the credit support annex **7**
9.1 Simulation . . . 7
9.2 Variation margin and the margin period of risk . . . 7
9.3 Initial margin . . . 8
9.4 Residual exposure . . . 8
9.5 Presets . . . 8
9.6 SA-CCR . . . 8

10 The XVA waterfall **8**

11 P&L attribution **9**
11.1 Independent price verification 9

12 Currency and digital-asset forwards **10**

13 FX options **11**

14 Market risk: ladder, VaR, stress, backtesting **13**
14.1 The ladder 13
14.2 Historical simulation by full revaluation 13
14.3 History and the demo's surfaces 14
14.4 Stress 14
14.5 Limits and backtesting 14

15 Counterparty credit: netting sets and real-world exposure **15**
15.1 Two measures 15
15.2 Joint simulation 15
15.3 Valuation on the paths, netting and the CSA 15
15.4 Measures and limits 16

16 Collateral regimes and counterparty capital **16**
16.1 Collateral classes and the two CSAs 16
16.2 SA-CCR by asset class (CRE52) 17
16.3 Capital, CVA capital, leverage 17
16.4 The book under regimes: attribution by channel 17

17 On-chain bilateral confirmation **18**
17.1 The canonical record (schema v2) 18
17.2 Signatures 18
17.3 The registry 18
17.4 Verification without Rijeka 19

18 Independent reference: QuantLib **19**

19 Model governance and known limitations **22**

1 Scope and principles

Rijeka is a front-to-back desk for over-the-counter derivatives: booking, pricing, cashflows, risk, XVA under a credit support annex, ISDA SIMM initial margin, and bilateral confirmation anchored to Ethereum. Phase 1 covers interest-rate products on overnight-index curves (USD SOFR, EUR €STR, GBP SONIA and others) and non-deliverable digital-asset forwards and options.

Three principles govern the implementation.

1. **No black boxes.** Each number is traceable to a formula in this document and a function in the repository; the Excel replications reproduce it outside the system.
2. **Risk taxonomy follows ISDA SIMM.** Sensitivities are `IR01` (all curves bumped), `IR01_DISC` (discount curves only), `THETA`, `GAMMA`, `VEGA`; the terms PV01 and DV01 are not used.
3. **Market data is dated and sourced.** Every snapshot carries a valuation date and a source. Pricing off a stale snapshot is permitted but is always flagged with the snapshot date; it is never silent.

1.1 Notation and conventions

t_0	valuation date
$P(t,T)$	discount factor from T to t on the OIS curve; $P(t_0,T)$ written $P(T)$
$\tau(d_1,d_2)$	year fraction under a day-count convention: ACT/360, ACT/365F, 30/360 bond basis (ISDA 2006 §4.16(f)), ACT/ACT ISDA
N, K, F	notional, strike, forward
σ_N	normal (Bachelier) volatility, quoted in basis points per annum
Φ, φ	standard normal distribution and density

Business-day adjustment is Modified Following on the currency's calendar unless the trade says otherwise. Spot lag is two business days for USD, EUR, JPY and CHF, one for CAD, zero for GBP, AUD and NZD (`pricing/bootstrap.py`, `CCY_SPOT_LAG`).

2 Discount curve construction

Module: `pricing/bootstrap.py`, `pricing/curve.py`.

2.1 Instruments

A curve is built from an overnight deposit and a strip of OIS par swap rates (tenors in weeks, months and years). Futures, FRAs and IRS quote types are accepted by the engine but Phase 1 curves use deposit and OIS swaps only.

2.2 Algorithm

1. The overnight deposit gives $P(t_0 + 1\text{d}) = 1/(1 + r_{ON}\,\tau_{ACT/360})$.
2. The spot discount factor $P(t_s)$ is obtained by extrapolating the overnight zero rate to the spot date and is *frozen* as an explicit pillar. This is the step that makes the bootstrap and the pricer agree to the last cent: both read the same $P(t_s)$.
3. Annual OIS swaps are bootstrapped on a dense integer-year grid (1Y to the longest quote, each year counted from the spot date, so a 29 February spot rolls to 28 February), with missing years filled by linear interpolation of par rates, so that every annual period end is a pillar. This gap fill is a convention: QuantLib's bootstrap of the same quotes interpolates log-linearly in discount factors between quoted maturities, and the two curves agree at the quotes but part by up to 1.3bp in zero rate (and 4bp in a single three-month forward) between them (§18).
4. Week- and month-tenor swaps are then bootstrapped in maturity order against the pillars already present.

For an OIS swap of par rate R with annual periods $(T_{i-1}, T_i]$, $i = 1 \dots n$, accrual factors τ_i (ACT/360 for USD), the terminal discount factor solves the par condition with the floating leg replaced by its telescoping value $P(t_s) - P(T_n)$:

$$P(T_n) \;=\; \frac{P(t_s) - R\sum_{i=1}^{n-1} \tau_i\, P(T_i)}{1 + R\,\tau_n}. \tag{1}$$

Because every period end is a pillar, a swap struck at its own par rate reprices to exactly zero.

2.3 Interpolation and extrapolation

Between pillar dates $d_0 < d < d_1$ the discount factor is log-linear in the calendar-day fraction $w = (d - d_0)/(d_1 - d_0)$:

$$P(d) = \exp\big[(1-w)\ln P(d_0) + w \ln P(d_1)\big], \tag{2}$$

equivalent to a piecewise-constant forward rate. Beyond the last pillar the continuously-compounded zero rate is held flat. Before the first pillar $P = 1$. The Excel replications implement exactly this rule with `MATCH/INDEX`, so a reader can check any discount factor the engine produces.

2.4 Risk bumps

A parallel shift of b basis points is applied to zero rates: $P_b(T) = P(T)\, e^{-b \cdot 10^{-4}\, T}$ with T in ACT/365F years (`Curve.shifted`).

3 Schedules and cashflows

Module: `pricing/schedule.py`, `pricing/day_count.py`, `pricing/ir_swap.py`.

Coupon periods are rolled from an anchor at the payment frequency (monthly, quarterly, semi-annual, annual, or zero-coupon), every date counted from the anchor (anchor $\pm\, k$ periods) so that a month-end does not drift: rolled back quarterly from 31 August the dates are 31 May, 28 February, 30 November, 31 August. No end-of-month rule is applied. The stub type chooses the anchor: `SHORT_FRONT` (the default) and `LONG_FRONT` roll backward from maturity, `SHORT_BACK` and `LONG_BACK` forward from the effective date, the long variants merging the odd period into its neighbour; explicit first-period-start and last-period-end dates override this. Period end dates are unadjusted; payment dates apply the business-day convention and the payment lag in business days of the leg's calendar (its `payment_calendar`, else its currency's: New York, TARGET, London, Tokyo, Zurich, Sydney, Toronto, or a join of them). Floating fixings are two business days before the period start on the same calendar. A period whose accrual ends on a weekend at month-end is therefore paid on the preceding business day, before its accrual ends; QuantLib, which adjusts accrual dates as ISDA does, would not build such a coupon, and the reference checks reproduce it coupon by coupon. (Before November 2026 the dates drifted after a short month, the stub type was ignored, and the lag and fixing dates counted weekdays only; NPV moved by less than 0.02bp of notional, individual coupon dates by up to three days.)

A fixed coupon over $(d_s, d_e]$ is $N\, r\, \tau(d_s, d_e)$. A floating coupon on a compounded overnight index is projected from the forecast curve as

$$r_{float} = \frac{1}{\tau(d_s, d_e)} \left(\frac{P_f(d_s)}{P_f(d_e)} - 1 \right) + s, \tag{3}$$

which equals the daily-compounded in-arrears rate implied by the curve; s is the spread. For a period already in progress, a realised fixing R (when supplied) covers the elapsed part τ_e and the curve's forward F the remainder τ_r, compounded: $r = \big[(1 + R\tau_e)(1 + F\tau_r) - 1\big]/\tau$. The realised part is rebuilt from the desk's own market data (`pricing/fixings.py`): the overnight rate is the ON quote of every daily snapshot of the leg's curve, carried over days without a snapshot and compounded on the index's basis (ACT/360 for SOFR, €STR and SARON, ACT/365F for SONIA and TONA); the price response names the snapshots it used. Without a snapshot on or before the period start the elapsed days are assumed to have accrued at the remaining stub's forward, and the response says so (the error is the realised-minus-forward gap times the elapsed fraction: 11bp of notional in the reference test, where they differ by 40bp). Notional, fixed-rate and spread schedules (step-ups, amortisation) are resolved per period from their effective dates.

Each coupon is discounted at its business-day-adjusted payment date, the date the cash moves. (Engine versions before October 2026 discounted at the unadjusted period end; where a date rolled over a weekend the two differ by a day or two of discounting. Corrected in v0.5, with the Excel replications as the independent check.)

4 Linear interest-rate products

Module: `pricing/ir_swap.py, pricing/greeks.py`.

The value of a leg is the signed sum of discounted coupons, $PV = \pm \sum_k C_k \, P(T_k)$, with pay legs negative. A swap's NPV is the sum of its legs; a basis swap is two floating legs on different forecast curves; an OIS swap is the single-curve case where forecast and discount curves coincide. The par rate is $R^* = PV_{float} \,/\, (N \sum_i \tau_i P(T_i))$.

Sensitivities are computed by bump-and-reprice:

- `IR01`: all curves shifted +1bp;
- `IR01_DISC`: discount curves shifted +1bp with forecasting held fixed (in single-curve mode this requires a separate bumped copy for discounting, so that `IR01_DISC` < `IR01`; the difference is the forecasting sensitivity);
- `THETA`: $NPV(t_0 + 1\text{d}) - NPV(t_0)$ with curves held;
- `GAMMA`: second difference of ±1bp shifts.

5 European swaptions and the volatility surface

Module: `pricing/swaption.py, pricing/sabr.py`.

5.1 Annuity, forward and the Bachelier price

For an option expiring at T_e into a swap paying annually at $T_1 < \cdots < T_n$ (dates rolled backward from the swap's maturity, with the trade's spot lag and business-day adjustment when the trade dates are supplied),

$$A = \sum_{i=1}^{n} \tau_i \, P(T_i), \qquad F = \frac{P(T_0) - P(T_n)}{A}. \tag{4}$$

Rates on overnight curves can be negative and are quoted in normal volatility, so the option is priced in the Bachelier model. With $v = \sigma_N \sqrt{T_e}$ and $d = (F - K)/v$,

$$V_{payer} = N \, A \, v \, \big[d \, \Phi(d) + \varphi(d) \big], \qquad V_{receiver} = N \, A \, v \, \big[- d \, \Phi(-d) + \varphi(d) \big]. \tag{5}$$

Greeks are produced by bump-and-reprice: vega per 1bp of σ_N, `IR01`, theta per calendar day, volga, vanna, dollar gamma, and a delta-hedge notional. A Hull–White cross-check price is reported alongside when the model is calibrated (§7).

5.2 Volatility tiers

The volatility used is tagged `QUOTED` (an at-the-money surface point at the trade's expiry and tenor), `SABR` (an off-the-money strike read from the calibrated smile), `INTERPOLATED` (between grid points), or `MANUAL` (user supplied). The tier travels with the price so that a valuation can be challenged.

5.3 $\beta = 0$ normal SABR

The smile at each expiry–tenor point is the normal SABR of Hagan et al. (2002) with backbone $\beta = 0$:

$$\begin{aligned} \sigma_N(K) &= \alpha \, C(F,K) \left\{ 1 + \left[\frac{2 - 3\rho^2}{24} \nu^2 + \frac{\rho \, \nu \, \alpha}{4} \right] T \right\}, \\ C = \frac{z}{\chi(z)}, \qquad z &= \frac{\nu}{\alpha}(F - K), \qquad \chi(z) = \ln \frac{\sqrt{1 - 2\rho z + z^2} + z - \rho}{1 - \rho}, \end{aligned} \tag{6}$$

with $C = 1$ at the money. Parameters (α, ρ, ν) are fitted to quoted volatilities at strike offsets $\{-200, -100, -50, -25, 0, +25, +50, +100, +200\}$bp by weighted least squares, with α constrained so the at-the-money volatility is matched exactly. The fit RMSE is stored with the parameters.

6 Caps, floors and collars

Module: `pricing/cap_floor.py, pricing/cap_vol_strip.py, pricing/cap_sabr.py.`

A cap on a compounded overnight index is a strip of quarterly caplets, ACT/360 accrual, from two calendar days after the valuation date, weekend dates moved to the following Monday. The rate fixes in arrears over the whole period; the caplet is valued with option time to the *middle* of its accrual period, $T = \left[(d_s - t_0) + \frac{1}{2}(d_e - d_s)\right]/365$ (`option_time`; documented as Bloomberg's convention). Time is on ACT/365F because normal vols are quoted per 365-day year; before November 2026 it was measured on ACT/360, which overstated the variance by 365/360. For caplet $(d_s, d_e]$ with forward F from (3) and normal volatility σ_N,

$$\text{caplet} = N\,\tau\,P(d_e)\left[(F-K)\,\Phi(d) + \sigma_N\sqrt{T}\,\varphi(d)\right], \qquad d = \frac{F-K}{\sigma_N\sqrt{T}}, \tag{7}$$

and a floorlet is the corresponding put. The at-the-money strike of a cap is the annuity-weighted average of its caplet forwards. A collar is a long cap and a short floor. QuantLib's compounded-overnight caplet pricer, which models the backward-looking accrual itself, values the reference caps 0.6–2.0% of premium lower and the reference floor 2.5% higher (§18); that is the size of the mid-period convention.

Booking. A cap, floor or collar is booked as one floating leg with zero participation in the rate (`leverage` 0) and the strikes on its `embedded_options` (a collar: a bought cap and a sold floor, or the reverse), so the swap pricer, the exposure engine and the P&L value exactly the caplet strip; the side, strikes, vol and premium are also in the trade's terms. A floating leg's leverage multiplies its projected rate everywhere (booked and ignored before November 2026). Until then these products were booked as a generic two-leg swap with no strike stored.

Caplet volatilities are stripped from flat cap volatilities one strike column at a time: the 1Y bucket is solved so that four Bachelier caplets reproduce the 1Y cap price; each later bucket (2Y, 3Y, 3–5Y, 5–7Y, 7–10Y) is solved after subtracting the value of the buckets already priced, piecewise flat within a bucket. A $\beta = 0$ SABR smile is then fitted per expiry bucket across strikes.

7 Bermudan swaptions: Hull–White trinomial tree

Module: `pricing/bermudan_swaption.py.`

The short rate follows $\mathrm{d}r_t = [\theta(t) - a\,r_t]\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$. Zero-coupon bonds are affine, $P(t,T) = A(t,T)e^{-B(t,T)r_t}$ with $B(t,T) = (1 - e^{-a(T-t)})/a$ (Brigo–Mercurio 2006, eq. 3.39). The tree is the Hull–White (1994) trinomial tree: a symmetric tree in $x_t = r_t - \alpha(t)$ with the standard branching rules, and a deterministic shift $\alpha(t)$ fitted at every grid time by Arrow–Debreu forward induction so that the tree reprices the market discount curve exactly. Backward induction values the underlying swap at each node by the single-curve telescoping identity $PV_{float}(t) = \sum_k N_k[P(t,T_{k-1}) - P(t,T_k)]$, and applies the early-exercise test at each exercise date.

Validation gate. Before any Bermudan value is trusted, the same tree must price European swaptions within 0.5% (at the money) and 1% (±100bp) of the closed-form Bachelier price

on a grid of expiries and tenors. **Scope.** The model is single-curve and single-tenor: multi-curve underlyings, capped or floored underlyings and basis underlyings are refused rather than approximated, and a Bermudan whose underlying tenor is far from the 5Y calibration column carries a warning.

8 Hull–White calibration for exposure simulation

Module: `pricing/calibration.py`.

With constant (a, σ), the model's at-the-money normal volatility for a swaption with expiry T_e on a swap with payment dates T_i, accruals α_i and annuity $A(0) = \sum_i \alpha_i P(0, T_i)$ is (Andersen–Piterbarg 2010, vol. 1 §4.4)

$$\sigma_N(T_e) = \sigma \, \frac{|w| \sqrt{V(T_e)}}{\sqrt{T_e}}, \qquad w = \sum_i \frac{\alpha_i P(0,T_i)}{A(0)} \, B(T_e, T_i), \qquad V(T_e) = \frac{1 - e^{-2aT_e}}{2a}, \tag{8}$$

where $P(0,T)$ is the *market* discount factor from the bootstrapped curve (a flat-rate approximation is the fallback only when no curve is supplied). A constant-parameter model cannot fit a whole surface; following standard XVA practice it is calibrated to the 5Y-tenor column (1Y×5Y, 2Y×5Y, 3Y×5Y, 5Y×5Y) by least squares, and the fit diagnostics are stored with the parameters. The calibration was verified against Jamshidian's decomposition to within 0.01%.

9 Exposure, collateral and the credit support annex

Module: `pricing/csa.py`, `pricing/xva_engine.py`, `api/routes/xva.py`.

9.1 Simulation

Short-rate paths are simulated under the calibrated Hull–White model on a monthly grid with the exact fit to the initial curve, and the trade is revalued on every path and date, giving mark-to-market paths $V(t)$ from our side. With $r(t) = f(0,t) + \phi(t) + x(t)$, x the driftless Ornstein–Uhlenbeck deviation and $\phi(t) = \frac{\sigma^2}{2a^2}(1 - e^{-at})^2$, a bond on a path is

$$P(t,T) = \frac{P(0,T)}{P(0,t)} \exp\Big[- B(t,T)\big(x(t) + \phi(t)\big) - \tfrac{\sigma^2}{4a} B(t,T)^2 \big(1 - e^{-2at}\big) \Big], \qquad B = \frac{1 - e^{-a(T-t)}}{a}. \tag{9}$$

The counterparty engine (`ccr/simulate.py`) omitted $\phi(t)$ until November 2026, which priced bonds high and put path forwards $\phi(t)$ below the curve (8bp at five years, 27bp at ten, for $a = 3\%$, $\sigma = 85$bp); the risk-neutral martingale check in the reference suite now pins it. Grid times are the grid dates' distance in 365.25-day years, the clock every trade date uses, so a fixing or expiry on a grid date sits on that grid point. A European swaption on the paths is valued before expiry with Bachelier on each path's forward swap rate at the normal vol the paths' own Hull–White model implies for it (the Jamshidian price at t_0 inverted through Bachelier, `ccr/options.hw_implied_vol`) and exercised at expiry on the same paths, so the value before expiry and the exercise are one model and expected exposure does not step at expiry (with today's market vol instead it stepped by the ratio of the two vols, 13–25% in the reference test, until November 2026); the market quote is reported beside the vol used. A sold option enters with the opposite sign (the terms' side). Embedded caplets fix on the path at their fixing date.

9.2 Variation margin and the margin period of risk

Variation margin held at t was called against the exposure one margin period of risk δ earlier:

$$C(t) = \max\big(V(t-\delta) - TH_{cp}, 0\big) - \max\big(-V(t-\delta) - TH_{own}, 0\big), \tag{10}$$

zeroed when $|C|$ is below the minimum transfer amount. The margin period δ is shorter than a grid step, so $V(t-\delta)$ is sampled from the Brownian bridge between grid points,

$$V(t-\delta)\,\big|\,V(t-\Delta t), V(t)\;\sim\;\mathcal{N}\big(V(t-\Delta t)+w\,\Delta V,\; w(1-w)\,s^2\big), \qquad w=\frac{\Delta t-\delta}{\Delta t}, \tag{11}$$

with s^2 the cross-path variance of ΔV. This preserves the variance of the move over the margin period; linear interpolation would understate it by a factor $\delta/\Delta t$.

9.3 Initial margin

Initial margin follows ISDA SIMM interest-rate delta: $IM = RW(\text{tenor}) \times |\texttt{IR01}|$ with the published risk weights for regular-volatility currencies (109bp at two weeks falling to 61bp at 5–20 years and 64bp at 30 years). The weights are calibrated to a 10-business-day margin period; for another period they scale by $\sqrt{\delta/10\text{bd}}$. Margin is assumed exchanged symmetrically and amortises with residual maturity. Received IM reduces exposure; posted IM is funded and drives MVA.

9.4 Residual exposure

$$E^+(t)=\max\big(V(t)-C(t)-IM_{recv}(t),\,0\big), \qquad E^-(t)=\min\big(V(t)-C(t)+IM_{post}(t),\,0\big). \tag{12}$$

Expected exposure $EE(t)=\mathbb{E}[E^+(t)]$, expected negative exposure $ENE(t)=\mathbb{E}[E^-(t)]$ and the 97.5% potential future exposure are reported per date.

9.5 Presets

Uncollateralised	no variation or initial margin.
Bilateral	daily VM, 10-business-day margin period (the BCBS–IOSCO floor for non-cleared trades), SIMM IM, thresholds and MTA from the CSA.
On-chain	atomic settlement: continuous VM, zero threshold and MTA, 5-business-day margin period (the cleared floor), SIMM IM scaled by $\sqrt{5/10}$.

The on-chain preset is the quantitative claim behind the confirmation layer: with continuous settlement the margin period halves, SIMM initial margin falls by $1-\sqrt{1/2}\approx 29\%$, and the SA-CCR maturity factor falls in the same proportion.

9.6 SA-CCR

For margined netting sets the maturity factor is $MF=1.5\sqrt{\max(\delta,\text{floor})/250}$ and replacement cost $RC=\max(V-C,\;TH+MTA-NICA,\;0)$ with $NICA$ the IM received; unmargined sets use $MF=\sqrt{\min(M,1)}$ and $RC=\max(V,0)$. The interest-rate supervisory factor is 0.5% and $\alpha=1.4$. Capital is 8% of risk-weighted exposure and feeds KVA.

10 The XVA waterfall

Module: `pricing/xva_engine.py`*, function* `run_waterfall`*.*

Default probabilities come from the credit triangle: hazard rate $h=s/LGD$ for a flat or term-structured spread s, survival $Q(t)=e^{-ht}$. With $D_i=P(t_i)$, Δt the grid step and

$\Delta Q_i = Q(t_{i-1}) - Q(t_i)$,

$$CVA = LGD_{cp} \sum_i EE_i \, \Delta Q_i^{cp} \, D_i \, \omega_i, \qquad DVA = -LGD_{own} \sum_i |ENE_i| \, \Delta Q_i^{own} \, D_i, \tag{13}$$

$$FVA = \sum_i s_i^{ftp} \, EE_i \, \Delta t \, D_i, \qquad FBA = -\sum_i s^{fba} \, |ENE_i| \, \Delta t \, D_i, \tag{14}$$

$$KVA = \sum_i \kappa \, K_i \, \Delta t \, D_i, \qquad MVA = -\sum_i IM_i \, \max(s_i^{ftp} - \pi, 0) \, \Delta t \, D_i, \tag{15}$$

where ω_i is an optional wrong-way-risk multiplier, κ the capital hurdle rate, K_i the SA-CCR capital on the exposure path, and π a yield pick-up on posted collateral (tokenised collateral that earns a return reduces MVA). Signs follow the convention that costs are positive. The waterfall is reported gross and under the CSA, so the benefit of collateral is visible term by term.

The charge on the ticket. Every ticket's XVA tab carries, under the trade's stand-alone number, the incremental charge: the counterparty's netting set valued on the credit engine's risk-neutral paths (§15) with the trade and without it, the same paths for both runs, the difference per component. That is the number a desk quotes; the stand-alone figure is the trade on its own under a chosen CSA preset. An opened or freshly booked trade shows the marginal form, the set with and without it. The route is `POST/api/xva/incremental`; the hypothetical trade is the ticket's booking body with its legs completed as the booking route would store them, so what is charged is what would be booked.

11 P&L attribution

Module: `pnl/explain.py, pnl/valuation.py`; full description in `docs/pnl_attribution.md`.

Daily P&L is attributed by sequential full revaluation: from yesterday's mark on yesterday's market the valuation is moved one thing at a time, in a fixed order (carry, cash, each curve tenor bucket by bucket, FX spot and points), each step's change a row, and what is left between the last step and today's mark on today's market is a *residual* row, shown and never spread. An amendment in the window is one step, $NPV(\text{new terms}, m_1) - NPV(\text{old terms}, m_1)$, after the market steps run on the old terms. The rows sum to the change in mark plus cash by construction. Cash counts the flows that left the mark, those paid in $[d_0, d_1)$: the pricer keeps a flow in the value on its payment date, so a coupon paying today is cash tomorrow (before November 2026 the window was $(d_0, d_1]$ and a payment date's P&L counted the coupon twice, reversed the next day). A volatility surface the trade is valued on (an FX option's) is moved after spot and points, bucket by bucket, as a VOL step; swaption and cap vols have no step yet and land in the residual. Booked European swaptions are valued as options in P&L, VaR, stress and the ladder (Bachelier at the day's nearest ATM grid vol; expiry, strike, payer or receiver and bought or sold from the booked terms, the fixed leg's forward start and rate when the terms are older); until November 2026 they were valued as their underlying swap.

11.1 Independent price verification

Module: `backend/ipv`; full description in `docs/ipv.md`. The controller verifies the inputs the book is priced on, not the trades: the list is the union over the trades in scope of the curves, funding levels, surfaces and spots they read (`pnl.valuation.trade\_curves`), and each input is laddered per pillar through the production pricers, one quote moved by one unit (1bp of a rate, 1bp of normal vol, one vol point, 1% of spot) and the dependent trades repriced. A pillar is a tenor, a swaption point (expiry × tenor) or an FX point (tenor, ATM/RR/BF); against an independent mark it carries a gap in the input's unit, a status against the firm's tolerance rule (the most specific rule wins) and a materiality flag ($|s| \cdot$ tol against the threshold). An

override is a market-data event with author and reason; its first-order P&L is $s\,\Delta$ and the full revaluation on the overridden market gives the residual as second order. Official P&L = flash + first order + second order + the day's netted XVA reserve change, signed by the controller once every input is settled and every trade valued; the trading P&L shows the flash beside it.

12 Currency and digital-asset forwards

Module: *pricing/currency_forward.py, pricing/crypto_forward.py, chain/canonical.py (CRYPTO_FORWARD, CRYPTO_OPTION).*

FX spot, forwards and non-deliverable forwards on a currency pair, and forwards on BTC, ETH or SOL against USD, are one product priced by covered interest parity (cash-and-carry): borrow the quote currency, buy the base today and lend it, so the forward must repay the quote-currency interest. Each currency discounts off its own overnight-index curve (USD SOFR, EUR €STR, GBP SONIA, JPY TONAR, CHF SARON, AUD AONIA, CAD CORRA); a digital asset discounts at its funding rate r_c (lending or staking yield, continuously compounded). With spot S in quote per base, $P_q(T)$ the quote-currency discount factor and $P_b(T)$ the base-currency one, and t_s the spot date (two business days of both currencies for a pair; the valuation date for a digital asset, whose quoted spot settles at once), $F = S\,\frac{P_b(T)/P_b(t_s)}{P_q(T)/P_q(t_s)}$ and $NPV_q = \pm N_b\,(F-K)\,P_q(T)$; quoted forward points are measured from the spot date. (Before November 2026 the spot was applied from the valuation date, dropping the carry over the spot lag: 5.6 pips on a six-month USDJPY from a Friday.) deliverable or non-deliverable changes the settlement mechanics, not the value. When a pair carries a quoted forward curve (forward points by tenor, saved like any other snapshot: Bloomberg, manual or import), the outright at a date is spot plus the points interpolated linearly in time, and the difference to the parity forward is reported as the cross-currency basis; the account chooses, in configuration, whether forwards are priced off quoted points or implied from the two curves, and every quote states which it used. For a digital asset against USD this reads

$$F = S\,\frac{e^{-r_c T}}{P_{USD}(T)} = S\,e^{(r_{USD}-r_c)T}, \qquad NPV = (F-K)\,N\,P_{USD}(T). \tag{16}$$

A non-deliverable European option is priced with Black-76 on that forward with lognormal volatility σ:

$$C = P_{USD}(T)\big[F\Phi(d_1) - K\Phi(d_2)\big], \qquad d_{1,2} = \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \tag{17}$$

with the put from parity. Settlement is in cash against a named settlement price source on a valuation date and time, and the canonical record carries exactly the fields the ISDA Digital Asset Derivatives Definitions name: digital asset, notional amount, strike or forward price and its currency, option type and exercise style, valuation and settlement dates, settlement currency and price source, premium amount, currency and payment date. **Status.** Implemented in the engine (`pricing/currency_forward.py`, with `pricing/crypto_forward.py` as the digital-asset case), priced from parameters and from booked trades on G10 pairs and BTC, ETH, SOL; spot from public reference rates (ECB) and a public exchange feed, the funding rate a desk input; the Excel replications rebuild the FX and the digital-asset forward cell by cell with the engine's NPV, spot delta (`FX01`, `CRY01`) and rate sensitivities (`IR01`; `FUND01` on a digital asset) beside the formulas. The canonical on-chain record carries an FX product (pair, both notionals, rate, value date, settlement type, NDF fixing fields) beside the digital-asset one. Options on the digital assets are the FX option of §13 with the funding rate as the base rate, cash-settled against the settlement price source the day after expiry; the trade window prices and books them in every style (2 November 2026). A validation note: the first workbook had the discount-factor ratio inverted (forward below spot at positive USD rates); the engine's parity test caught it and both were corrected on 28 October 2026.

13 FX options

Module: `pricing/fx_option.py`, `pricing/fx_vol.py`; *route* `POST/api/currency/option/price`; *booked valuation in* `pnl/valuation.py`, *exposure in* `ccr/options.py`, *record in* `chain/canonical.py`.

A European option on a currency pair BASE/QUOTE is the right to buy (call) or sell (put) the base currency against the quote currency at the strike K, exercised at the expiry time on the expiry date and settled on the settlement date, two business days later in both currencies' calendars, by delivery of both currencies or, for a non-deliverable option, in the quote currency against a fixing. It is valued with Garman and Kohlhagen (1983), which is Black–Scholes with the base currency as the dividend-paying asset, on the same forward the currency forward uses (§12: covered interest parity from the spot date, or the account's quoted points), discounted in the quote currency to the settlement date:

$$C = N_b\,P_q(T_s)\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big], \qquad P = N_b\,P_q(T_s)\big[K\,\Phi(-d_2) - F\,\Phi(-d_1)\big], \qquad d_{1,2} = \frac{\ln(F/K) \pm \tfrac{1}{2}\sigma^2 T_e}{\sigma\sqrt{T_e}}, \tag{18}$$

with T_e the ACT/365 time to expiry and T_s to settlement, F the forward to settlement and N_b the base notional. A sold option carries the opposite sign. The premium is a cash flow on its payment date, not part of the value once paid; a non-deliverable option has the same value as a deliverable one.

Volatility. A pair's surface is saved as the market quotes it, per tenor: the at-the-money vol, the 25-delta risk reversal $RR = \sigma_{25\Delta c} - \sigma_{25\Delta p}$ and the 25-delta butterfly $BF = \tfrac{1}{2}(\sigma_{25\Delta c} + \sigma_{25\Delta p}) - \sigma_{ATM}$. Across strikes the smile is Malz's quadratic in the call's forward delta,

$$\sigma(\Delta) = \sigma_{ATM} - 2\,RR\,(\Delta - \tfrac{1}{2}) + 16\,BF\,(\Delta - \tfrac{1}{2})^2, \tag{19}$$

which reproduces the three quotes exactly and is solved as a fixed point with $\Delta = \Phi(d_1)$ (the premium-excluded forward delta). ATM, RR and BF are interpolated linearly in time between quoted tenors and held flat beyond them; a tenor quoted ATM-only contributes to the level and not to the skew. Conventions stated as limitations: the ATM quote is applied at $\Delta = \frac{1}{2}$, so the vol read at $K = F$ sits a hundredth of a point off it; the wings beyond 10-delta are the quadratic extrapolated; premium-included deltas (the standard for USD-base pairs) are not used. Without a saved surface the pair prices at a flat desk default and the response says so; the daily demo build writes a synthetic surface for the G10 pairs, labelled as such, since no public source quotes FX vols.

Risk. In the desk's taxonomy: `spot01` (+1% spot, repriced; the forward moves with spot, the discounting does not), the equivalent spot position in base units, gamma over a further 1%, vega per volatility point (repriced), `IR01` on each curve (+1bp, repriced: the quote curve raises the forward and lowers the discount factor, the base curve lowers the forward), theta one calendar day later. The ticket shows the forward delta beside them.

In the frameworks. The P&L explain gains a VOL step after spot and points: the pair's surface is moved tenor bucket by tenor bucket, so a day's vega P&L is a row of its own and only cross terms reach the residual (§11). The risk ladder bumps each surface bucket by one volatility point and reports vega by (surface, bucket), kept apart from the rate ladder (`docs/market_risk.md`); Volatility is a VaR factor: each surface's quotes (ATM, risk reversal and butterfly per tenor; a swaption's ATM grid point by point) are moved by their observed daily changes beside the curves and spots and every option is repriced in full; a surface is secondary to the curves and spots, held still on a day it has no observation, and the page says how many of the window's days each surface covered. The hosted demo's surfaces have no public source, so their level is scaled day by day by the realised vol of the series they sit on,

labelled as such (`risk/vols.py`). On the exposure paths the option is valued before expiry with Garman–Kohlhagen on the path's forward (the path's spot and bonds) at the surface's forward vol for its remaining life and the smile at its moneyness, and after expiry as the payoff fixed on the spot at the expiry grid point and discounted until settlement (§9, §15). The canonical record (§17) carries the fields of the ISDA FX and Currency Option Definitions: pair, option type, exercise style, both notionals, strike, expiry date and time, settlement date and type, settlement currency and fixing source for a non-deliverable option, premium amount, currency and payment date, automatic exercise; buyer and seller are named by LEI so both bookings hash identically.

Styles. (`pricing/fx_exotic.py`.) The same dynamics price the other styles a currency desk trades, each with its QuantLib benchmark:

- *American*: a Leisen–Reimer binomial tree (801 steps) with early exercise at every node; within 0.02bp of notional of QuantLib's QD+ fixed-point engine and to rounding of its own Leisen–Reimer tree.
- *Digital*: a fixed quote-currency amount (cash-or-nothing) or a fixed base amount (asset-or-nothing scaled), Reiner–Rubinstein closed forms; to rounding against `AnalyticEuropeanEngine`.
- *Barrier* (up or down, in or out, with a rebate paid at the hit for a knock-out and at expiry for a knock-in): the knock-out value is the integral of the payoff against the survival density of log-spot with one absorbing barrier, built by the method of images with the drift restored by Girsanov, and knock-in = vanilla − knock-out; within 2×10^{-3}bp of notional of `AnalyticBarrierEngine` across all eight cases.
- *Double barrier*: the same integral with the two-barrier image series (Ikeda–Kunitomo); within 2×10^{-3}bp of `AnalyticDoubleBarrierEngine`.
- *Touch*: one-touch and no-touch from the first-passage and survival probabilities (paid at the hit or at expiry), double no-touch from the two-barrier density, double one-touch as its complement; against `AnalyticDigitalAmericanEngine`, its knock-out variant and `AnalyticDoubleBarrierBinaryEngine`.

Barriers are monitored continuously; the vol is the smile vol at the strike, held flat across the barrier (vanna–volga is the next step; QuantLib's analytic barrier engines are flat-vol as well). On the exposure paths a barrier is monitored at the monthly grid points, which understates knock probabilities a little, and an American option is valued as its European (the early-exercise premium is not simulated); both are said in `docs/ccr.md`. The vol along the paths is the surface's, not one number held: the spot steps with the piecewise forward vols of the ATM term structure, so its log-variance at every grid date is the surface's ATM variance there, and at each date the option is valued at the forward vol for its remaining life with the quoted risk reversal and butterfly at the remaining tenor read at the delta of the path's own forward against the strike (sticky-delta, the FX convention); a touch, which has no strike, takes the forward ATM vol. The same functions (`pricing/fx\_vol.py`: `step\_vols`, `path\_vols`) serve the ticket's stand-alone XVA and the credit engine. Vol of vol is not modelled: the surface's level does not move along a path, which a stochastic-vol model would add and is on the roadmap with its own validation gate. The canonical record carries the style, the payout and its currency, the barrier type and levels, the rebate, the touch type, whether a touch pays at the hit, and the monitoring convention. The ticket's scenario tab reprices every style on a grid of spot and vol shocks through the engine itself. The trade's structure in the blotter is its style, so a digital, a barrier and a touch are told apart at a glance and found by search.

Digital assets. An option on BTC, ETH or SOL against USD is the same product with the asset's funding rate r_c as the base rate: the forward is the cash-and-carry one of §12, $b = r_{USD} - r_c$, discounting in USD to settlement the day after expiry, settlement in cash against the settlement price source (a non-deliverable option in the ISDA Digital Asset Derivatives Definitions' vocabulary; the canonical record is the `CRYPTO_OPTION` product, which keeps its

pre-November byte shape for a European and carries the style's terms otherwise). Every style of §13 prices on the same engine; the reference table carries rows at crypto parameters (spot 65 000, funding 3%, vol 60%). The desk's taxonomy names the sensitivities: `FX01` is the spot delta of a currency pair and `CRY01` that of a digital asset (PV for +1% spot); on a digital asset `IR01` is the discounting curve's sensitivity alone and the funding rate's is `FUND01`, kept apart on the ticket and in the ladder, whereas on a currency pair both legs are rate curves and `IR01` is their sum. The surfaces `\{ASSET\}USD\_VOL` have the FX shape; no public source quotes crypto implied vols the hosted demo can rely on, so their level is scaled day by day by the realised vol of the public spot series like the FX surfaces, labelled as such, and a venue-implied surface (Deribit) is the next step. The Methodology tab's workbooks rebuild the European option and the styles at crypto parameters beside the engine's values. **Reference.** Against QuantLib's `GarmanKohlagenProcess` with `AnalyticEuropeanEngine` on the same spot, rates and vol: value to rounding for calls and puts, bought and sold, in and out of the money, to two years; delta in base units to rounding; the repriced spot01, vega and rate sensitivities within the second- and third-order terms of their bumps (§18). The smile reproduces its three quotes at their own strikes. The styles above are in the same table. **Replication.** The Methodology tab's workbooks rebuild the European option (Garman–Kohlhagen with its delta and vega), the digitals, the eight single-barrier cases with rebates (Reiner–Rubinstein's terms A–F), the touches, and the double barriers and double touches (the two-barrier image series summed one row per image) with live Excel formulas beside the engine's values: the closed forms are the independent check of the engine's numerical integration, and agree with it to better than 10^{-6} of value. The American option is a tree and has no spreadsheet form; its check is QuantLib's.

14 Market risk: ladder, VaR, stress, backtesting

Modules: *`risk/sensitivities.py`, `risk/var\_full.py`, `risk/var.py`, `risk/stress.py`, `risk/backtest.py`, `risk/history.py`, `risk/vols.py`; full description in `docs/market\_risk.md`.*

14.1 The ladder

For every live trade and every curve it is valued on, each tenor bucket's quotes are bumped +1bp and the trade repriced on the as-of market through the production pricers; the result is key-rate IR01 by (curve, bucket), the buckets (0–1Y, 1–5Y, 5–10Y, 10Y+) the same the P&L explain uses. Spot deltas are the pricers' PV for +1% spot: `FX01` on a currency pair, `CRY01` on a digital asset. On a digital asset `IR01` is the discounting curve's sensitivity alone and the funding rate's is `FUND01`, reported beside the IR01 totals and never summed into them. A volatility surface the trade is valued on (`\{PAIR\}\_VOL`, `\{CCY\}\_SWVOL\_ATM`) is bumped +1 vol point per tenor bucket, a swaption grid by expiry, and reported as vega by (surface, bucket). A par bump of every quote is a slightly smaller move in zero rates, so the ladder total sits about 2% below the ticket's IR01, which bumps zero rates; the bucket values agree with a QuantLib rebootstrap to 0.05% of DV01 (§18).

14.2 Historical simulation by full revaluation

The risk factors are the curve quotes (par rates in basis points, tenor by tenor), the spots (log returns) and the volatility surfaces (every quote of an FX surface: ATM, 25-delta risk reversal and butterfly per tenor; the swaption ATM grid point by point). For each day i of the window today's market M is shifted by that day's observed changes Δ_i and every trade repriced:

$$\mathrm{P\&L}_i = \sum_{\text{trades}} \big[NPV(M{+}\Delta_i){-}NPV(M)\big]\cdot fx, \qquad \mathrm{VaR}_q = -Q_q\{\mathrm{P\&L}_i\}, \qquad \mathrm{ES}_q = -\mathbb{E}[\mathrm{P\&L}_i \mid \mathrm{P\&L}_i < -\mathrm{VaR} \tag{20}$$

the horizon scaled by $\sqrt{h}$. The curves and spots define the window: only days on which every one of them has an observation are used, and a factor with no history is listed on the page, not dropped. A volatility surface is a secondary factor: on a day it was observed its quotes move with the rest, on a day it was not it is held still, and the page states each surface's coverage of the window. The bootstrapped curves are cached per shifted market; beyond a budget of valuations the full revaluation is skipped and the ladder VaR reported, with a note. The ladder VaR is computed alongside as the linear comparison,

$$\mathrm{P\&L}_i^{\mathrm{ladder}} = \sum_{c,b} \mathrm{KR01}[c,b]\,\Delta_i[c,b] + \sum_{a} \mathrm{spot01}[a]\,r_i[a] + \sum_{s,b} \mathrm{vega}[s,b]\,\Delta\sigma_i[s,b], \qquad (21)$$

and the difference between the two is the convexity and the inside-bucket curve shape the ladder cannot see. Defaults: confidence 99% (95% secondary), one day, a 730-day window, sixty observations minimum.

14.3 History and the demo's surfaces

Risk-factor history comes from the public sources the demo prices on: Treasury par yields and NY Fed SOFR, ECB €STR and the AAA par curve, Bank of England SONIA and gilt yields, ECB reference FX rates, Coinbase daily closes; each day's curve is built with the live spread schedules and stored as the template account's snapshot, so the daily rebuild extends the series. No public source quotes implied volatility, so the demo's surfaces are synthetic in shape and market-driven in level: each day the parametric surface's ATM is scaled by $\sqrt{\sigma^{\mathrm{real}}_{\text{tenor window}}/\sigma^{\mathrm{real}}_{\text{series}}}$ from the stored spot series (an FX pair, a digital asset) or from the 5Y par rate's changes (the swaption grid), clipped to $[\frac{1}{2}, 2]$, risk reversals and butterflies as quoted; every such row is labelled. A desk with a feed stores its own surfaces under the same ids.

14.4 Stress

Named scenarios by full revaluation: every trade repriced on the shifted market. Rates scenarios shift quotes in basis points, parallel or by bucket (parallel ±25, ±100bp; steepener and flattener ±25 short against long); FX scenarios move the non-USD asset by a percentage, a USD-first quote the other way (USD ±10%); digital assets ±30%; a combined risk-off (rates −50bp, USD +5%, digital assets −30%); and the VOL scenarios move the FX surfaces' ATM rows by a number of points (larger for the digital assets) and the swaption grid by bp of normal vol. A rate scenario leaves the surfaces alone. A trade whose pricer cannot value the shifted market is listed under the scenario, not summed as zero.

14.5 Limits and backtesting

A limit names a metric (`VAR\_99\_1D`, `ES\_99\_1D`, `IR01`, `IR01:<curve>`, `KR01:<curve>:<bucket>`, `SPOT01:<asset>`, `NPV`, `STRESS:<scenario>`), a scope and a value; utilisation is $|value|/limit$ with a warning level and a breach at 100%, evaluated on every member's book after the daily run. Backtesting pairs every stored P&L date with the VaR standing the evening before; an exception is a day whose loss exceeded it, on the clean (positions held at the previous close), hypothetical (clean without carry) or dirty P&L. The count is read two ways: the Basel traffic light from the cumulative binomial at the stated confidence (green below 95%, yellow to 99.99%, red beyond: at 250 days and 99%, 0–4 / 5–9 / 10+), and Kupiec's proportion-of-failures test, $LR = -2\ln[(1-p)^{n-x}p^x] + 2\ln[(1-x/n)^{n-x}(x/n)^x] \sim \chi^2_1$, two-sided, so too few exceptions is also a finding. Sixty paired days are needed before the page says anything.

Known limitations. One window, equally weighted (no filtered historical simulation yet); swaption and cap vol moves are in VaR through the ATM grid only, and cap vols have no P&L step (§11); the smile's own moves enter the ladder VaR through vega × ATM only.

15 Counterparty credit: netting sets and real-world exposure

Modules: `ccr/simulate.py`, `ccr/valuation.py`, `ccr/options.py`, `ccr/calibrate.py`; *full description in* `docs/ccr.md`.

15.1 Two measures

The same model is run under two parameter sets and every number is labelled with its measure. *Risk-neutral*: the Hull–White parameters of §9 calibrated to the swaption surface, each spot stepping with the ATM term structure of its pair's surface (a desk default, labelled, for a pair without one), the drift from the curves; its outputs are the XVA reserves and belong to the P&L. *Real-world*: volatilities measured from two years of public history (daily changes of each currency's 3M rate, log returns of each spot), a mean reversion set by policy ($a = 0.05$; two years of daily data cannot identify it, and the estimate is shown beside it), zero drift by default (the forward slope is a risk-neutral artefact, the historical drift the firm's alternative), and the correlation of all factor shocks from the same history; its outputs are EE, EPE, PFE and the limits.

15.2 Joint simulation

One Hull–White factor per currency and one lognormal factor per spot on a monthly grid to the netting set's last maturity, Brownian motions correlated by the calibrated matrix (identity in the risk-neutral run). Grid times are each grid date's distance from the as-of date in 365.25-day years, the clock every trade date is read on. The short rate is $r(t) = f(0,t) + \varphi(t) + x(t)$ with x the driftless Ornstein–Uhlenbeck deviation and $\varphi(t) = \frac{\sigma^2}{2a^2}(1 - e^{-at})^2$; bonds are affine in $x + \varphi$ and reproduce today's curve,

$$P(t,T) = \frac{P(0,T)}{P(0,t)} \exp\Big(-B(t,T)\,(x_t+\varphi(t)) - \frac{\sigma^2}{4a} B(t,T)^2 (1-e^{-2at})\Big), \qquad B(t,T) = \frac{1 - e^{-a(T-t)}}{a}. \tag{22}$$

(Until November 2026 the exponent was evaluated on x_t alone, which priced every bond high by $e^{B\varphi}$: 8bp at five years, 27bp at ten for $a = 3\%$, $\sigma = 85$bp; the martingale check $\mathbb{E}[D(t)P(t,T)] = P(0,T)$ in the reference suite pins it.) Spots follow $S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t)$, μ the rate differential or zero. **Historical sampling** is the default real-world measure: each monthly step is a block of 21 consecutive observed daily changes from the calibration window, the block's start drawn uniformly and the same block for every factor; the block's sum is the step's increment, in the Hull–White recursion $x_i = x_{i-1}e^{-a\Delta t} + \sum \Delta r$ in place of $\sigma\sqrt{\Delta t}\, z$, and the step's log return for a spot; the panel is demeaned so the drift is the stated one. This keeps the window's fat tails, its volatility clustering within a block and the realised co-movement of rates and spots without a correlation matrix; Gaussian shocks are the alternative, chosen per firm, and the fallback when a factor has no history on common dates.

15.3 Valuation on the paths, netting and the CSA

Every trade of a counterparty is valued on the same paths and summed per path and date, so netting is exact: swaps from the path's bonds, forwards as $\pm N(S_t\, DF_b/DF_q - K)\, DF_q$. Options are on the paths too: a caplet is Bachelier on the path's forward with today's ATM cap vol held flat until its fixing, then the realised payoff discounted to payment; a European swaption is Bachelier on the path's forward swap rate at the normal vol the paths' own Hull–White model implies for it (the Jamshidian price inverted through Bachelier, so value before expiry and exercise at expiry are one model and exposure does not step at expiry), then the underlying swap where it was in the money (physical settlement); an FX or digital-asset option is valued on the path's forward at the surface's forward vol for its remaining life and the smile at its

moneyness (sticky-delta), its payoff fixed at the expiry grid point and discounted to settlement; barriers and touches carry which paths have knocked, monitored at the grid points seen so far; an American option is valued as its European. Anything else is listed as not simulated, never valued at zero. The CSA is the engine of §9: variation margin against the value lagged by the margin period of risk (a Brownian bridge inside the step), initial margin when the record says so, residual $E^+ = \max(V - C - IM, 0)$.

15.4 Measures and limits

$EE(t)$ the mean of E^+; $PFE_q(t)$ its q-quantile over paths at 95, 97.5 and 99%; EPE the time average of EE over the first year; the peak PFE_{95} and its date; the largest PFE_{95} in each tenor bucket; current exposure $\max(V_0 - C_0, 0)$. Credit limits are set on `PFE\_PEAK:<cpty>`, `PFE:<cpty>:<bucket>`, `EPE:<cpty>`, `CE:<cpty>` and `CVA:<cpty>` with the same utilisation and breach rules as the market-risk limits. The reserves come from the risk-neutral paths through the waterfall of §9 with the counterparty's spread (a desk default of 85bp and 40% LGD until the record carries its own).

Known limitations. The bridge noise is drawn independently per step, so the 99% tail is jagged month to month while the 95% profile is smooth; one factor per currency, so no curve twist in the exposure; the surface's level does not move along a path (its term structure and smile are applied, §13; vol of vol is not simulated); the risk-neutral spot paths of a pair without a surface use a desk default vol, labelled; wrong-way risk and stressed EEPE are on the roadmap.

16 Collateral regimes and counterparty capital

Modules: `pricing/collateral.py`, `pricing/sa\_ccr.py`, `pricing/capital.py`, `ccr/saccr\_map.py`; full description in `docs/capital.md`. Built from the public texts named at each formula; no third-party material.

16.1 Collateral classes and the two CSAs

A CSA (§9) says whether collateral is exchanged; a *regime* says what it is and what each rulebook makes of it. A collateral class carries its recognition under the credit-risk-mitigation rules (CRE22.34–22.45; SCO60.94 for tokenised and crypto assets), its supervisory haircut by residual maturity (CRE22.49), its eligibility as initial and variation margin (CFTC §23.156), its liquidity level (LCR30) and its cryptoasset group (SCO60). A regime is a CSA, an eligible set, the compositions posted and received, and three labelled switches: close-out enforceability, the initial-margin eligibility state of a conditional class, and the margin period of risk (the non-cleared floor or the cleared floor for atomic settlement). From one regime the engine derives two CSAs:

$$\text{regulatory:}\quad C_{reg} = C\,\big(1 - h\big),\quad h = 1 - \textstyle\sum_i w_i \,\mathrm{rec}_i\,(1 - h_i),\quad \text{margined iff recognised VM is exchanged and en} \tag{23}$$

$$\text{economic:}\quad C_{eco} = C \text{ when enforceable, else } 0 \text{ (gross exposure)}, \tag{24}$$

with w_i the composition weights, $\mathrm{rec}_i \in \{0, 1\}$ the recognition and h_i the haircut. The regulatory CSA feeds SA-CCR, CVA capital and the leverage exposure; the economic CSA feeds CVA, DVA, FVA, FBA and MVA, with the pickup on posted margin the yield of its eligible part. The all-recognised regime reproduces the plain CSA field by field, so every number of §9 is the cash-and-government-securities column.

16.2 SA-CCR by asset class (CRE52)

$$EAD = 1.4\,(RC + m \cdot AddOn), \qquad m = \min\Big(1,\ 0.05 + 0.95\,e^{(V-C)/(1.9\,AddOn)}\Big), \tag{25}$$

$$RC = \max(V - C, 0) \text{ unmargined}, \qquad RC = \max(V - C,\ TH + MTA - NICA,\ 0) \text{ margined}, \tag{26}$$

$$d_i = N_i\,SD_i, \quad SD_i = \tfrac{e^{-0.05S_i} - e^{-0.05E_i}}{0.05}, \quad MF_i = \sqrt{\min(M_i, 1)} \text{ or } 1.5\sqrt{\delta/250}, \tag{27}$$

(CRE52.1, 52.10, 52.18–52.19, 52.23, 52.34, 52.48, 52.52). The supervisory delta is ±1 for linear trades and $\pm\Phi\big(\pm(\ln\frac{P+\lambda}{K+\lambda} + \frac{1}{2}\sigma^2 T)/(\sigma\sqrt{T})\big)$ for options at the supervisory volatility (CRE52.40; λ the rate shift). Interest-rate add-ons are per currency over three maturity buckets, $EN = \sqrt{D_1^2 + D_2^2 + D_3^2 + 1.4D_1D_2 + 1.4D_2D_3 + 0.6D_1D_3}$ at 0.5% (CRE52.57); FX per pair $|\sum \delta\, d\, MF|$ at 4% (CRE52.59); the cryptoasset class per asset–fiat pair at 32% without supervisory duration (SCO60.98); credit, equity and commodity through the single-factor formula with the Table 2 parameters (CRE52.72). Classes add without diversification (CRE52.25); a margined set is capped at the same set unmargined (CRE52.2). The netting set maps to rows as `docs/capital.md` states (a physically settled swaption on the forward par rate, active to the underlying's end; forwards by the foreign leg in USD; options at the supervisory delta of spot and strike). Along the exposure path the KVA leg calls the same functions with the dates shifted, so the snapshot and the profile are one computation; with a single interest-rate trade it equals the $SF \cdot N \cdot SD \cdot MF$ of §9.

Reference. The Basel Committee's worked examples of March 2014 (Annex 4a, examples 1–5) are reproduced unrounded in `tests/test\_sa\_ccr.py`: EAD 569 (three rate trades, swaption delta −0.27), 381 (credit, multiplier 0.965), 5 406 (commodities, nine-month maturity factor), 936 (combined) and 1 879 (margined weekly, MPoR 14 days, independent amount 150, RC 0, multiplier 0.958).

16.3 Capital, CVA capital, leverage

$RWA = RW \cdot EAD$ and $K_{CCR} = 8\%\,RWA$ with the standardised risk weight by counterparty type and grade (sovereigns CRE20.7, banks CRE20.18, corporates CRE20.42–20.43; broker-dealers as banks when prudentially supervised, else as corporates, CRE20.40; an explicit override on the record, labelled). CVA capital follows the reduced basic approach,

$$SCVA_c = \tfrac{1}{1.4}\,RW_c\,M\,EAD\,DF(M), \quad DF(M) = \tfrac{1-e^{-0.05M}}{0.05M}, \qquad K_{CVA} = 0.65\sqrt{(0.5\textstyle\sum_c SCVA_c)^2 + 0.75\sum_c} \tag{28}$$

(MAR50.14–50.15) with RW_c by sector and grade (MAR50.16, Table 1). The waterfall's KVA becomes $-\kappa\sum_i \big(K_{CCR}(t_i) + K_{CVA}(t_i)\big)\Delta t\, D_i$ with $M(t)$ the remaining maturity; without a CVA weight it is the KVA of §9 exactly. The leverage exposure is $1.4\,(\max(V - C_{cash}, 0) + AddOn)$ with only cash variation margin reducing it (LEV30); the HQLA value of the margin posted uses the Level 1, 2A and 2B haircuts (LCR30). The cost difference between two regimes is quoted as a running spread over the trade's annuity or an upfront spread of notional.

16.4 The book under regimes: attribution by channel

Every netting set is simulated once on the risk-neutral paths and re-collateralised step by step from the CSA as booked to an on-chain CSA under today's rules (zero threshold and MTA, CRE52.18–52.19; daily margining, CRE52.50; the dispute doubling removed, CRE52.51; recognised collateral and enforceable netting, CRE22.34 and SCO60.94; eligible initial margin in the asset held at its yield, CFTC §23.156), then to the forward-looking states (the cleared-floor margin period of risk; a tokenised sovereign claim eligible as initial margin; Level 1 liquidity).

Each step changes one thing in the previous step's CSA and the difference it makes is that channel's attribution, sequential as the P&L explain of §11; the steps sum to the columns exactly. The record's margin period of risk is $9 + N$ business days for re-margining every N days, floored at 10 and doubled while disputed. A record already on cash and government securities shows zero in the collateral step (parity, not superiority); SIMM initial margin moves only with the margin period of risk; market-risk capital is not a function of any step. *Module: ccr/capital_report.py; route /api/capital/portfolio.*

Known limitations. Initial margin is a delta margin per risk class without the cross-class aggregation, vega or curvature; supervisory deltas are held fixed along the path; the leverage and liquidity figures are the trade's stock, not the ratios; the cryptoasset initial-margin weight is a labelled assumption.

17 On-chain bilateral confirmation

Modules: chain/canonical.py, chain/signing.py, chain/attestation.py; the registry contract chain/src/TradeConfirmationRegistry.sol; the verifier chain/verify/index.html.

17.1 The canonical record (schema v2)

Both parties sign one record that either can derive alone from its own booking. The record is keyed by the unique transaction identifier, names the two parties by LEI in sorted order, names the payer and receiver LEI of every leg, the buyer and seller of every embedded option and the payer and receiver of every custom cashflow, and orders legs by content. It contains only the economic terms: no internal references, no book or desk, no status, no timestamps, no party names. Serialisation rules, which a verifier in any language must reproduce exactly: JSON, UTF-8, keys sorted at every level, separators `,` and `:` with no whitespace; numbers as decimal strings without exponent, trailing zeros or negative zero; dates ISO-8601; nulls and booleans preserved; strings unchanged. The trade hash is $h = \text{keccak256}(\text{bytes})$. The schema version is inside the payload so a confirmation is always re-derived under the version it was signed with.

17.2 Signatures

Each party signs an EIP-712 typed message in the domain `RijekaTradeConfirmation`, version 1, bound to the chain and the registry address:

```
TradeConfirmation(bytes32tradeHash,addresscounterparty)
TradeAmendment(bytes32prevHash,bytes32newHash,addresscounterparty)
TradeTermination(bytes32tradeHash,addresscounterparty)
```

Each party's message names the *other* party's address, so a signature cannot be replayed against a different counterparty, chain or registry. A firm's signing address is registered in Rijeka through a wallet challenge; once registered, Rijeka can verify that firm's signatures but can never produce them. On the hosted demo Rijeka holds keys for demo entities only, and any confirmation it signs for both sides is marked `bilateral:false`.

17.3 The registry

The registry contract has no owner and no administrative function. `confirm` verifies both signatures over the confirmation digest and records $(h, \text{partyA}, \text{partyB}, \text{timestamp})$ with status *Confirmed*; `amend` verifies both signatures over the amendment digest, marks the previous hash *Superseded* and records the new hash with the previous one as its parent; `terminate` verifies

both signatures and sets *Terminated*. Signatures with high s values are rejected. Once a trade is anchored, Rijeka refuses off-chain edits to its terms: an amendment is itself a bilateral act.

17.4 Verification without Rijeka

A proof pack contains the canonical record, the hash, both signatures and the anchor. The public verifier, a single self-contained page, (i) re-serialises the record and recomputes the keccak256 hash, (ii) rebuilds both EIP-712 digests and recovers both signers, (iii) reads the registry on a public node and checks status and parties. No Rijeka server is contacted. The contract source is verified on Sourcify so the reader can audit what the registry does before trusting it. **What it proves:** that these exact terms were signed by these two keys and recorded at that block. **What it does not prove:** that a key belongs to a legal entity; that binding is a matter of the firms' own key management and registration.

18 Independent reference: QuantLib

Suite: `backend/tests/reference/`*; table:* `python -m tests.reference.report`.

Every number above is rebuilt in QuantLib (version 1.44), an independent open-source library, on a frozen USD SOFR snapshot of 33 quotes with the shape of the daily demo build (valuation date 30 October 2026). The suite runs in continuous integration with every commit; each comparison carries a tolerance and fails the build when the difference grows past it. Two kinds of comparison are made.

- **Replica (same conventions).** QuantLib is given Rijeka's discount factors (a `DiscountCurve`, log-linear on ACT/365F, which is Rijeka's interpolation) and Rijeka's stated conventions, and must reproduce Rijeka's number to rounding. Schedules are generated by QuantLib's own `Schedule`, so a date error shows as a value difference; SOFR is projected by an overnight index fixing every calendar day, which reproduces $P(d_s)/P(d_e) - 1$ exactly; Hull–White comparisons use QuantLib's `HullWhite` model with the same (a, σ).
- **Convention.** QuantLib's own bootstrap and market SOFR conventions (`OISRateHelper`: payment lag two days, modified following, the SOFR fixing calendar), QuantLib's compounded-overnight caplet pricer, or a different but legitimate definition of a sensitivity. These differences are reported with their cause and bounded.

The inputs are a public-data curve rebuild and synthetic volatilities: a clean comparison says the engine computes what the methodology says, not that the demo market is a live mark.

Product	Metric	n	Worst	Unit	Tol.	Kind
Curve	implied rate	1	-2.62e-10	bp	1.00e-06	replica
Curve	par rate on Rijeka DFs	32	2.80e-11	bp	1.00e-06	replica
Curve	zero rate (cc, ACT/365F)	39	1.25	bp	1.5	convention
Curve	simple fwd ACT/365F, replica	7	-8.84e-12	bp	1.00e-06	replica
Curve	simple fwd ACT/365F, market bootstrap	7	4.3	bp	6	convention
IRS	NPV, replica	15	-1.12e-11	bp of notional	1.00e-06	replica
IRS	DV01 (+1bp zero), replica	12	1.17e-10	% of DV01	1.00e-04	replica
IRS	NPV, QuantLib market conventions	12	-0.0974	bp of notional	1	convention
IRS	par rate, replica	4	-4.00e-13	bp	1.00e-06	replica
IRS	NPV vs true accrual	1	-11.1	bp of notional	12	convention
FX forward	forward rate	3	-2.22e-12	pips	1.00e-06	replica

Product	Metric	n	Worst	Unit	Tol.	Kind
FX forward	NPV (quote ccy)	3	-3.07e-12	bp of notional	1.00e-06	replica
FX forward	spot01 (+1% spot)	3	1.52e-12	% of value	1.00e-06	replica
FX forward	IR01 quote curve	3	1.90e-10	% of value	1.00e-06	replica
FX forward	IR01 base curve	3	4.51e-10	% of value	1.00e-06	replica
FX forward	forward from the spot date	3	-2.22e-12	pips	1.00e-06	replica
FX forward	NPV, spot date T+2 (joint calendar)	3	2.13e-12	bp of notional	1.00e-06	replica
FX forward	NPV with no spot date vs T+2 market	3	4.6	bp of notional	5	convention
Crypto NDF	forward price	3	0	USD	1.00e-06	replica
Crypto NDF	NPV (USD)	3	-1.97e-12	bp of notional	1.00e-06	replica
Crypto NDF	NPV vs replication by hand	3	-3.74e-13	bp of notional	1.00e-06	replica
FX option	NPV (USD)	10	-2.64e-12	bp of notional	1.00e-06	replica
FX option	delta (base units) vs QuantLib delta × N	5	-1.91e-12	% of delta	1.00e-06	replica
FX option	spot01 (+1% repriced) vs delta·ΔS + ½gamma·ΔS²	5	-0.411	% of spot01	1	convention
FX option	vega (+1 vol pt repriced) vs QuantLib vega × 0.01	5	1.2	% of vega	2	convention
FX option	ir01 quote curve vs QuantLib rho × 1bp	5	8.22e-05	% of value	0.02	convention
FX option	ir01 base curve vs QuantLib dividendRho × 1bp	5	1.05e-04	% of value	0.02	convention
FX option	NPV vs QuantLib (discounted to expiry)	1	0.484	bp of notional	0.5	convention
FX smile	smile vol at the quote's own strike	4	-3.35e-09	vol points	1.00e-06	replica
FX smile	forward delta at that strike	4	8.01e-11	delta	1.00e-06	replica
FX smile	smile vol at the forward vs the ATM quote	2	0.0301	vol points	0.05	convention
FX smile	linear in time between 3M and 1Y	1	0	vol points	1.00e-09	replica
FX option styles	NPV vs QuantLib QD+ fixed point	4	-0.0128	bp of notional	0.02	convention
FX option styles	NPV vs QuantLib Leisen–Reimer tree, 801 steps	4	-3.67e-09	bp of notional	1.00e-05	replica
FX option styles	NPV	8	-1.40e-11	bp of payout	1.00e-06	replica
FX option styles	NPV vs QuantLib AnalyticBarrierEngine	32	1.63e-04	bp of notional	0.002	replica
FX option styles	NPV vs QuantLib AnalyticDoubleBarrierEngine	8	1.05e-05	bp of notional	0.002	replica
FX option styles	NPV vs QuantLib AnalyticDigitalAmericanEngine	4	-3.28e-09	bp of payout	0.002	replica
FX option styles	NPV vs QuantLib AnalyticDigitalAmericanKOEngine	2	3.28e-09	bp of payout	0.002	replica
FX option styles	= payout × DF	4	0	bp of payout	1.00e-06	replica
FX option styles	NPV vs QuantLib AnalyticDoubleBarrierBinaryEngine	2	1.16e-12	bp of payout	0.002	replica
Crypto option	NPV (USD) vs Black-76 on the carry forward	2	-3.36e-13	bp of notional	1.00e-06	replica

Product	Metric	n	Worst	Unit	Tol.	Kind
Crypto option	NPV vs QuantLib cash-or-nothing	1	-1.46e-12	bp of payout	1.00e-06	replica
Crypto option	NPV vs QuantLib Reiner–Rubinstein	1	-1.93e-04	bp of notional	0.002	replica
Crypto option	NPV vs QuantLib digital American	1	0	bp of payout	0.002	replica
Crypto option	NPV vs QuantLib double-barrier binary	1	1.91e-12	bp of payout	0.002	replica
Swaption	NPV	21	9.31e-13	bp of notional	1.00e-06	replica
Swaption	vega (1bp vol)	6	2.99e-14	% of vega	1.00e-06	replica
Swaption	delta: analytic forward delta vs +1bp curve bump	6	13.8	% of delta	15	convention
Cap/floor	NPV, replica	4	4.39e-06	% of premium	0.001	replica
Cap/floor	vega (1bp vol), replica	4	-6.99e-04	% of vega	0.05	replica
Cap/floor	NPV vs QuantLib compounded-SOFR caplet pricer	4	-2.48	% of premium	3	convention
Cap/floor	NPV through the swap pricer, replica	3	-5.95e-08	% of premium	1.00e-06	replica
Bermudan (HW1F)	Bermudan NPV, tree vs QuantLib tree	12	-0.428	bp of notional	1	replica
Bermudan (HW1F)	European NPV, tree vs QuantLib Jamshidian	12	-0.444	bp of notional	1	replica
European (HW1F)	Jamshidian closed form, Rijeka vs QuantLib	12	4.51e-05	bp of notional	0.001	replica
CCR paths	martingale (bp of price)	10	-0.623	standard errors	4	replica
CCR paths	E[D·V] at expiry vs QuantLib Jamshidian	12	-1.07	standard errors	4	replica
CCR paths	value today (Bachelier at the HW implied vol) vs QuantLib Jamshidian	3	-0.338	bp of notional	0.5	replica
CCR paths	HW implied vol vs the market vol it replaces	3	-17.9	bp	40	convention
CCR paths	E[D·payoff] vs QuantLib HW zero-bond put	3	-1.13	standard errors	4	replica
Key-rate ladder	KR01 (+1bp quotes, rebootstrap)	20	0.0314	% of DV01	0.5	convention
Key-rate ladder	ladder total (par +1bp) vs ticket IR01 (zero +1bp)	5	-2.14	% of DV01	3.5	convention

Causes of the convention differences. Curve: the integer-year gap fill (§2). Swaps: market accrual adjustment and payment lag against Rijeka's default unadjusted accrual and the booked lag; under 0.1bp of notional. FX: the spot-date carry, now applied (the row keeps the old, valuation-date convention visible). Swaption delta: the ticket shows both the analytic forward delta $A\,\Phi(d)\cdot 1\text{bp}$ with the annuity held and the curve DV01 (a +1bp parallel zero shift, repriced), which also moves the annuity; up to 14% apart on a 5Y×10Y. Risk and P&L use the curve bump. Caps: the mid-period option time against QuantLib's compounded-rate model (§6). Exposure: the Hull–White implied vol a swaption is valued at on the paths against the market quote (§9); the value today against QuantLib's Jamshidian price is within 0.35bp of notional, the paths reading the curve at one-day granularity. Ladder: a 1bp move in par quotes is a slightly smaller move in zero rates, so the ladder total sits about 2% below the ticket's `IR01`; both are published and labelled.

Findings fixed in November 2026, each pinned by a test: the counterparty path bond formula (S1); booked swaptions valued as their underlying swap in P&L and market risk (S1); coupon dates drifting after a short month, and the stub type ignored (S3, dates off by up to three days); the curve failing to build when spot is 29 February (S2); zero-coupon legs failing to price (S2); the FX forward applied from the valuation date instead of the spot date (S2, up to 3.7bp of notional); cap option time on ACT/360 (S2, +0.7% of an ATM premium); a period under way adding realised and forward SOFR instead of compounding them (S3); the P&L cash window counting a coupon twice on its payment date (S2); payment lag and fixing dates in weekdays and New York business days only (S3); an unreachable legacy FX pricer with an inverted discount-factor ratio, removed (S4). **Fixed on 2 November 2026:** realised overnight fixings for periods under way, from the daily snapshots (§3); option terms stored at booking, caps and floors as zero-participation legs, sold options with their sign (§6); a float leg's leverage honoured; caplets discounted at the payment date; the swaption's forward delta labelled with the curve DV01 beside it; the exposure engine's swaption vol made consistent with its exercise (§9). **Added on 2 November 2026:** FX options (§13): European, American, digital, barrier, double barrier and touch, each against its QuantLib engine, with the VOL step in the P&L explain and vega on the ladder. **Also on 2 November 2026:** volatility surfaces as VaR factors and the swaption ATM grid in the explain's VOL step and on the vega ladder; digital-asset options in every style on the FX option engine (§13), with `CRY01`, `IR01` as the discounting sensitivity and `FUND01` named throughout. **Open:** cap vol moves have no P&L step and land in the residual; swaptions and caps booked before November 2026 carry no option terms and are valued from their legs.

19 Model governance and known limitations

- **Validation.** Versioned validation reports live in `model_validation/`; v0.4.2 covers IRS and OIS (38 tests against Bloomberg), later versions swaptions (delta of $3 on a $10m 5Y), caps (0.1% at matched volatility), SABR and the scenario engine.
- **Replications.** The Excel workbooks in the Methodology tab are regenerated on request from the user's own snapshot. Every book has the same shape: a cover with the trade and the checks' verdict; the ticket as the trader saw it, rebuilt in cells, every number a link into the sheet that derives it; the derivation, line by line, each quantity with its live value and its cell; the inputs, the curve and the product's own sheets; a scenario chart for the options; and a Checks sheet with every formula against the engine's value, a tolerance and a PASS or FAIL that the cover sums up. Plain spreadsheet features only, so the file opens clean in Excel, Numbers and Google Sheets. The XVA book (§9, `methodology/xva\_workbook.py`) goes further: it carries the engine's own random draws, rebuilds every Hull–White path, the bond reconstitution, the margin-period bridge, the collateral and the waterfall in cells, shows the swaption calibration with the model's closed-form vol per basket point and an RMSE grid around the fit, and has a LIVE mode that re-rolls the simulation in the spreadsheet with a convergence sheet for the Monte Carlo error. The FX and digital-asset XVA books (`methodology/fx\_xva\_workbook.py`) do the same for a forward: the lognormal paths from the engine's draws, the pair's surface interpolated to the forward's tenor with the smile fixed point at $K = F$ shown iteration by iteration, the initial margin from the spot sensitivity and the pair's weight, and the same CSA, bridge, waterfall and LIVE mode.
- **Independent reference.** Every product is repriced in QuantLib on every commit (§18); a difference that grows past its stated tolerance fails the build.
- **Open items, stated plainly.** Hull–White is single-curve and calibrated to one tenor column (§7–8). SABR is $\beta = 0$ only. The hosted demo prices off a dated snapshot with no live feed, and says so on every screen. The registry contract has had no external audit. SIMM covers interest-rate delta; vega and other risk classes are on the roadmap.
- **Change control.** Every change to a pricing module is a commit in a public repository

with its tests; this document is versioned with the code.

References

- Andersen, L. and Piterbarg, V. (2010). *Interest Rate Modeling*, vols. I–III. Atlantic Financial Press.
- Brigo, D. and Mercurio, F. (2006). *Interest Rate Models: Theory and Practice*, 2nd ed. Springer.
- Hagan, P., Kumar, D., Lesniewski, A. and Woodward, D. (2002). Managing smile risk. *Wilmott*, September, 84–108.
- Hull, J. and White, A. (1994). Numerical procedures for implementing term structure models I. *Journal of Derivatives* 2(1), 7–16.
- Gregory, J. (2020). *The xVA Challenge*, 4th ed. Wiley.
- Green, A. (2015). *XVA: Credit, Funding and Capital Valuation Adjustments*. Wiley.
- BCBS (2014). *The standardised approach for measuring counterparty credit risk exposures* (SA-CCR).
- BCBS–IOSCO (2015). *Margin requirements for non-centrally cleared derivatives*.
- ISDA (2024). *ISDA SIMM Methodology*, version 2.6; ISDA (2023). *Digital Asset Derivatives Definitions*.
- Ethereum Improvement Proposal 712: Typed structured data hashing and signing.